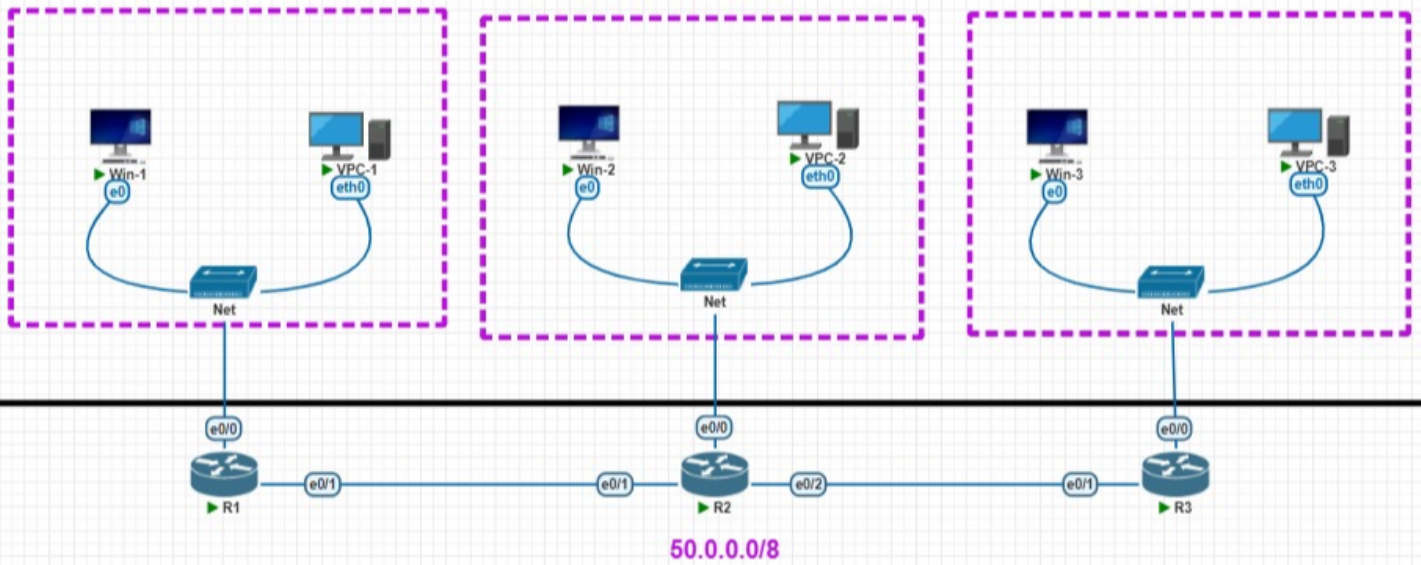


## SUBNETTING

HR 50-PC

PRODUCTION 120-PC

SALES 250-PC



 R1

```
R1(config-if)#exit
R1(config)#interface ethernet0/1
R1(config-if)#ip address 50.0.1.193 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#ip route 0.0.0.0 0.0.0.0 50.0.1.194
R1(config)#exit
R1#
*Aug 12 18:01:37.819: %SYS-5-CONFIG_I: Configured from console by console
R1#ping 50.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 50.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R1#ping 50.0.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 50.0.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R1#
*Aug 12 18:02:41.811: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip dhcp pool HR
R1(dhcp-config)#network 50.0.1.128 255.255.255.192
R1(dhcp-config)#default-router 50.0.1.129
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#lease 1 2 3
R1(dhcp-config)#exit
R1(config)#ip dhcp pool PRODUCTION
R1(dhcp-config)#network 50.0.1.0 255.255.255.128
R1(dhcp-config)#default-router 50.0.1.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#lease 1 2 3
R1(dhcp-config)#exit
R1(config)#ip dhcp pool SALES
R1(dhcp-config)#network 50.0.0.0 255.255.255.0
R1(dhcp-config)#default-router 50.0.0.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#lease 1 2 3
R1(dhcp-config)#exit
R1(config)#
```

 R2

```
*Aug 12 17:45:30.979: unix_ll_sock_notify: Invalid L1 socket file descriptor fd
= 6
```

```
*Aug 12 17:45:30.979: unix_ll_sock_notify: Invalid L1 socket file descriptor fd
= 6
```

```
*Aug 12 17:45:30.979: unix_ll_sock_notify: Invalid L1 socket file descriptor fd
= 6
```

```
*Aug 12 17:45:37.073: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
0, changed state to up
```

```
*Aug 12 17:45:37.074: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
1, changed state to up
```

```
*Aug 12 17:45:37.074: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to up
```

```
*Aug 12 17:45:37.074: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to up
```

```
R2#configure terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
R2(config)#i
```

```
*Aug 12 17:45:48.135: %LINK-3-UPDOWN: Interface Ethernet0/3, changed state to do
wn
```

```
*Aug 12 17:45:49.135: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to down
```

```
R2(config)#interface ethernet0/0
```

```
R2(config-if)#ip address 50.0.1.1 255.255.255.128
```

```
R2(config-if)#no shutdown
```

```
R2(config-if)#exit
```

```
R2(config)#interface ethernet0/1
```

```
R2(config-if)#ip address 50.0.1.194 255.255.255.252
```

```
R2(config-if)#no shutdown
```

```
R2(config-if)#exit
```

```
R2(config)#interface ethernet0/2
```

```
R2(config-if)#ip address 50.0.1.197 255.255.255.252
```

```
R2(config-if)#no shutdown
```

```
R2(config-if)#exit
```

```
R2(config)#
```

```
*Aug 12 17:50:37.803: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Confi
g Wizard)
```

```
R2(config)#ip route 50.0.0.0 255.255.255.0 50.0.1.198
```

```
R2(config)#ip route 50.0.1.128 255.255.255.192 50.0.1.193
```

```
R2(config)#
```

 R3

```
R3#
R3#
R3#
R3#
R3#
R3#configure te
*Aug 12 17:42:33.406: %LINK-3-UPDOWN: Interface Ethernet0/2, changed state to do
wn
*Aug 12 17:42:33.411: %LINK-3-UPDOWN: Interface Ethernet0/3, changed state to do
wn
*Aug 12 17:42:34.406: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to down
*Aug 12 17:42:34.411: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to down
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#interface ethernet0/0
R3(config-if)#ip address 50.0.0.1 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#interface ethernet0/1
R3(config-if)#ip address 50.0.1.198 255.255.255.252
R3(config-if)#no shutdwon
      ^
% Invalid input detected at '^' marker.

R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#exit
R3#show
*Aug 12 17:44:33.700: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip interface brief
Interface                IP-Address      OK? Method Status          Prot
ocol
Ethernet0/0              50.0.0.1        YES manual up              up
Ethernet0/1              50.0.1.198      YES manual up              up
Ethernet0/2              unassigned      YES TFTP  down            down
Ethernet0/3              unassigned      YES TFTP  down            down
```

